

First Last

+1 6767676767 | firstlast@gmail.com | [LinkedIn](#) | [GitHub](#) | U.S. Citizen | City, State

EDUCATION

T10 CS University

Bachelor of Science in Computer Science, GPA: 4.0

City, State

Aug 2023 - May 2027

Clubs & Activities: ICPC, University Trading Club

Relevant Coursework: Computer Organization & Programming, Machine Learning, Data Structures & Algorithms, Combinatorics, Differential Equations, Machine Learning, Design & Analysis of Algorithms

EXPERIENCE

Oracle (Cloud Infrastructure)

August 2024 – May 2025

Software Engineer Intern

City, State

- Developed Python- and Terraform-based automation to configure global OCI virtual networking components (VCNs, route tables, gateways) across **10+ regions**, enabling consistent, repeatable network deployments at scale.
- Automated CI/CD workflows for network configuration validation & rollout, implementing pre-deployment checks and staged releases that **reduced network misconfiguration incidents by 25%** during regional expansions/upgrades.
- Worked closely with network engineers and SREs (8-12 engineers) to support **low-latency network services**, **improving visibility into network changes & accelerating safe deployment** of routing and connectivity updates.

University Research Group

May 2024 – July 2024

FPGA Engineer Intern

City, State

- Designed RF signal processing pipelines in SystemVerilog** for real-time data acquisition, achieving **sub-50ns end-to-end latency & processing 250 MSPS throughput** across **4-channel parallel architecture**.
- Optimized RTL modules for timing closure at **200+ MHz**, **reducing resource utilization by 25%** through pipelining and algorithmic improvements while **meeting sub-100ps jitter requirements for signal integrity**.
- Collaborated with software and RF engineers to **integrate FPGA logic with C++ host interface**, delivering reliable hardware system tested across **500+ hours of continuous operation**.

University Lab

May 2023 – July 2023

Undergraduate Research Assistant

City, State

- Built ML pipeline **processing 1,000+ acoustic samples**, applying **FFT-based feature extraction and spectral filtering** in NumPy to generate 50+ frequency-domain features for supervised learning.
- Trained a Random Forest classifier using scikit-learn, experimenting with ensemble sizes (**50-500 trees**) and max-depth constraints, **achieving 93% test accuracy with 0.91 F1-score** and validating model robustness through bootstrapped confidence intervals.
- Performed systematic feature selection via SHAP values and permutation importance, identifying dominant spectral bands (**2-8 kHz**) **driving 85% of predictive power**, reducing feature set from 50 to 35 dimensions and **improving training efficiency by 2.5x**.

PROJECTS

Lab Work Project | FPGA, RTL, HLS

Jan 2026 – Present

- Developed a simulation-driven optimizer for FPGA dataflow designs, tuning FIFO buffers to reduce on-chip memory usage by up to **85%** while preserving performance, with **10⁵ + faster evaluation than RTL co-simulation**.

TECHNICAL SKILLS

Languages: Python, C/C++, SystemVerilog, Java, Bash, Linux

Libraries: NumPy, Pandas, scikit-learn, SHAP, Matplotlib

Skills: RTL Design, HLS Dataflow, OCI, AWS (EC2, EMR, Redshift), Terraform

Interests: Basketball, Football, Poker, Star Wars